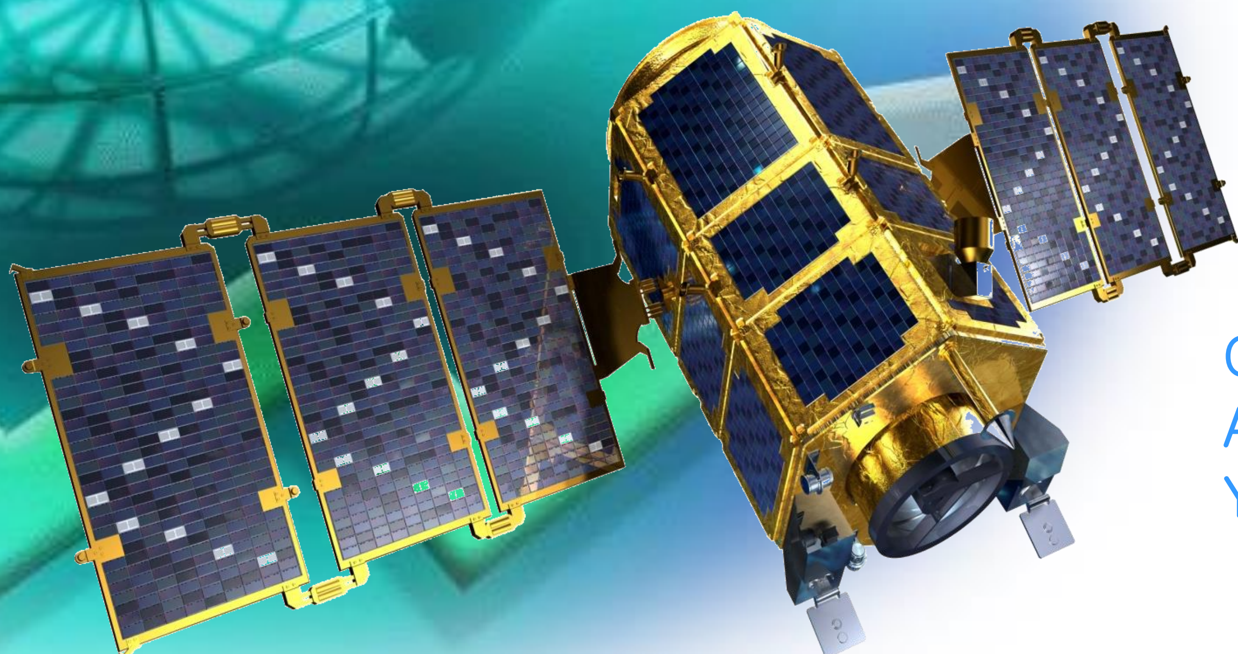
Fall 2011

Spacecraft Systems



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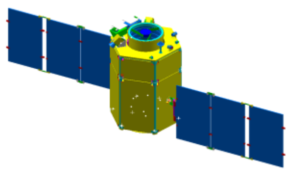
Chapter 10
SPACECRAFT DESIGN & SIZING

10.1 Requirements, Constraints, and the Design Process
10.2 Spacecraft Configuration
10.3 Design Budgets
10.4 Designing the Spacecraft Bus
10.5 Integrating the Spacecraft Design
10.6 Examples

- **To design a S/C requires us to understand the mission characteristics, including:**
  - Payload's size and characteristics
  - S/C system constraints: orbit, lifetime, operations, etc.

- **Then configure a spacecraft to:**
  - Carry the payload equipment
  - Provide the functions necessary for mission success



An unmanned S/C consists of at least three elements

- Payload
  - Mission-peculiar equipment or instruments
- S/C Bus
  - Carries the payload
  - Provides house keeping functions
    - Support the payload mass
    - Point the payload correctly
    - Keep the payload at the right temperature
    - Provide electric power, commands, telemetry
    - Put the payload in the right orbit and keep it
    - Provide data storage & communication
- Booster Adapter
  - Provides the load-carrying interface with the booster
  - Booster: First-stage of multi-stage launch vehicle or strap-on rocket used to augment the core launch vehicles' takeoff thrust and payload capability

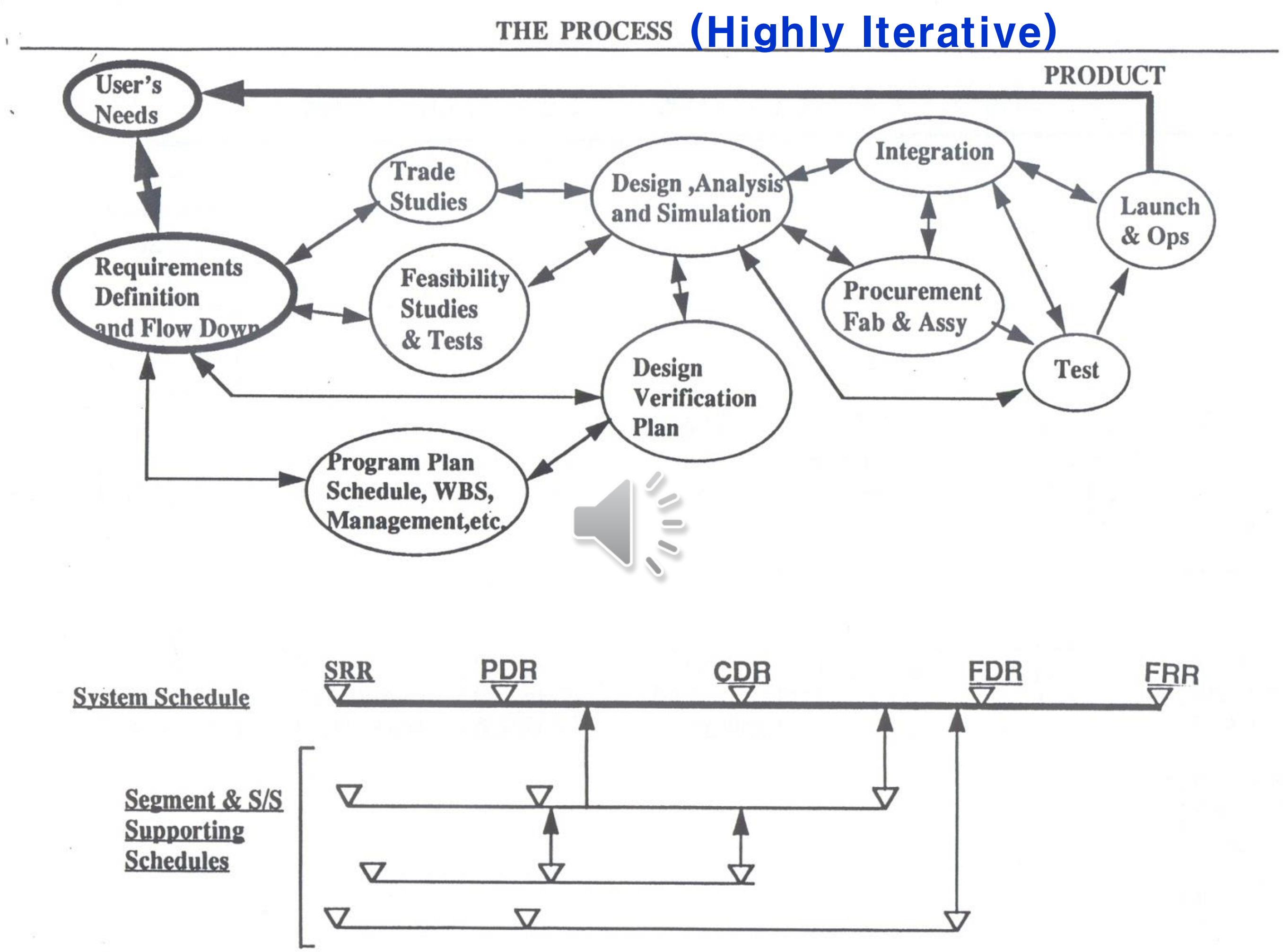
◆Overview of S/C Design and Sizing (Table 10-1)

TABLE 10-1. Overview of Spacecraft Design and Sizing. The process is highly iterative, normally requiring several cycles through the table even for preliminary designs.

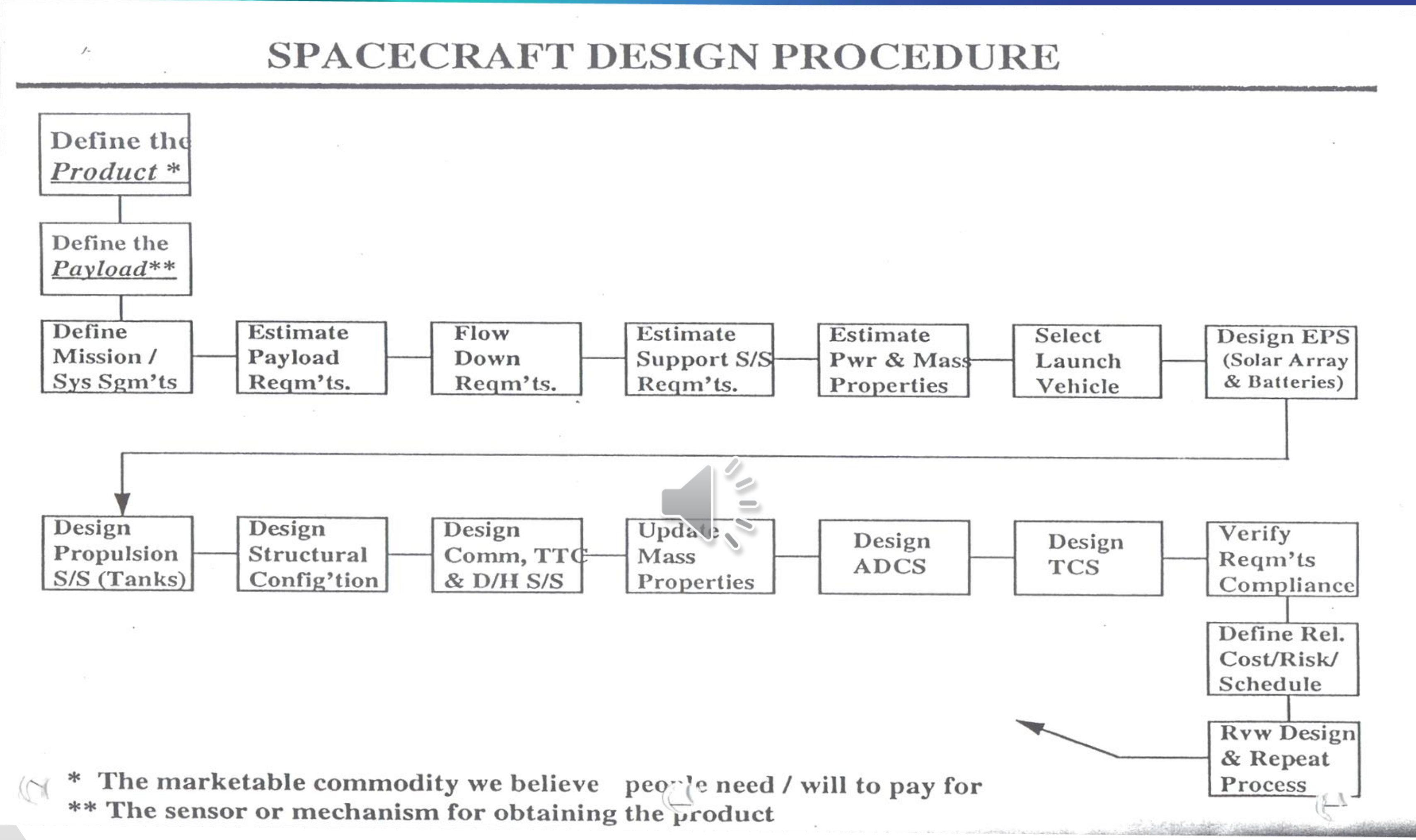
Step	References
1. Prepare list of design requirements and constraints	Sec. 10.1
2. Select preliminary spacecraft design approach and overall configuration based on the above list	Sec. 10.2
3. Establish budgets for spacecraft propellant, power, and weight	Sec. 10.3
4. Develop preliminary subsystem designs	Sec. 10.4
5. Develop baseline spacecraft configuration	Secs. 10.4,10.5
6. Iterate, negotiate, and update requirements, constraints, design budgets	Steps 1 to 5

1. **Requirements and Constraints are dictated by:**
   - Mission Concept
   - Mission Architecture
   - Payload Operation

2. **Select preliminary S/C design approach and overall configuration based on the above list**
   - Payload Mass ⇒ S/C의 Mass
     - Size, Shape ⇒ Size
     - Power ⇒ Power
     - Weight ⇒ Size
   - S/C Power ⇒ Solar Array Size ⇒ Solar Array Type
   - Booster Diameter ⇒ S/C Diameter
   - Pointing Requirement ⇒ S/C Body Orientation



- SRR: System Req's Review
- PDR: Prelim. Design Review
- CDR: Critical Design Review
- FDR: Final Design Review
- FRR: Flight Readiness Review

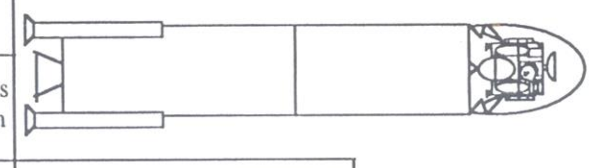


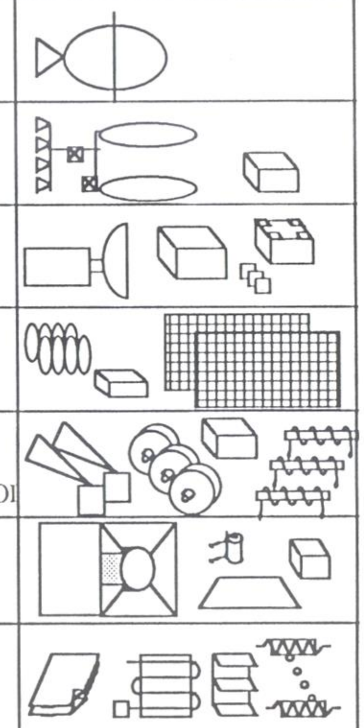
- S/S: Subsystem
- EPS: Electrical Power System
- TTC: Tracking, Telemetry and Command
- D/H: Data Handling
- ADCS: Attitude Determination & Control System
- TCS: Thermal Control System

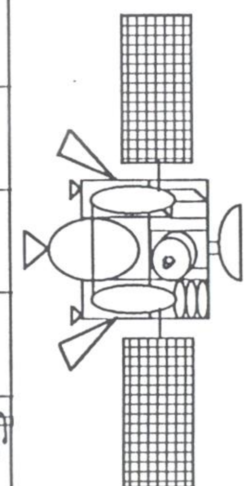
REQUIREMENTS TRACEABILITY & SPACECRAFT DESIGN

(Read Matrix Counter Clockwise*)

Mission Reqmts									
Orbit Parmtrs	**L/V**								SC Mass & Geom
Orbit Parmtrs	Transfer Orbit	**AKM**							Initial Mass
			RCS			Maneuver & Mntm Dump Rq			
r, freq S/N,BER GSD,etc.		TM/Cmd Rq	TM/Cmd Rq	**P/L & TT&C**	TM/Cmd & Comp Rq	TM/Cmd & Comp Rq	TM/Cmd & Comp Rq	TM/Cmd & Comp Rq	
Orbit Parmtrs	Elect I/F Rq	Power Profile Volt Rq	Power Profile Volt Rq	Power Profile Volt Rq	**EPS**	Pwr Prfl Volt Rq Att Prfl	Power Profile Volt Rq	Power Profile Volt Rq	
Orbit Parmtrs & Targets		Spin Rq	Isp, Thrust Level & Δt min	Pntg, Jitter & Nat Frq	Pntg & Nat Frq	**ADCS**	RCS Lever Arms		Mass center MOI/POI
	Mech I/F Rq	Algnmt /volume Rq	Algnmt /volume Rq	Algnmt /volume FOV Rq	Algnmt /volume FOV Rq	Algnmt /volume FOV Rq	**SMS**	Algnmt /volume FOV Rq	
Orbit Parmtrs		Pwr Loss Temp Rq		Pwr Loss Temp Rq	Pwr Loss Temp Rq	Pwr Loss Temp Rq Att Prfl	Pwr Loss Temp Rq	**TCS**	
	Perform-ance, Rel, $ & Sch	Isp, Mass/ Geom,Rel $ & Sch	Isp, Mass/ Geom, Rel $ & Sch	Mass/ Geom, Rel $ & Sch	Mass/ Geom, Rel $ & Sch	Mass/ Geom, Rel $ & Sch	Mass/ Geom, Rel $ & Sch	Mass/ Geom, Rel $ & Sch	**Mass Pr Rel'blty $ & Sch**







∫ Test, Mgmt and Design Verification

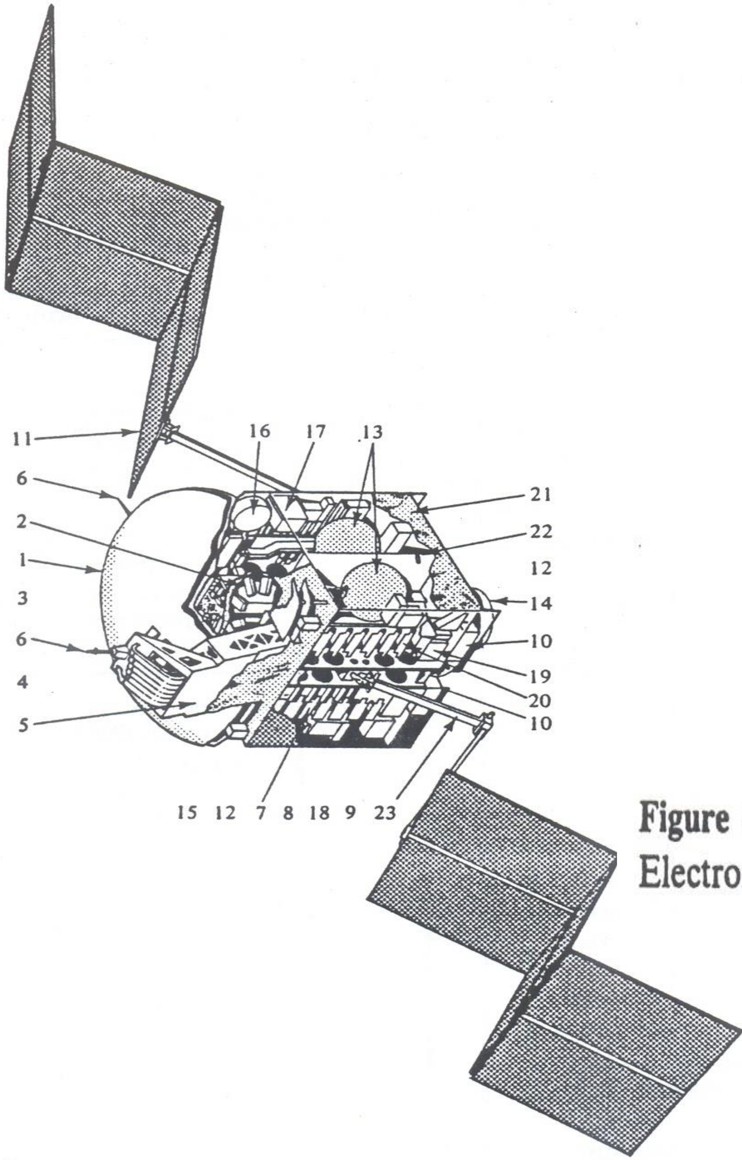
Performance, MMD Cost & Schedule

* Columns=flow of requirements from "diagonal". Rows=flow of requirements to "diagonal".

◆ Spacecraft Subsystem (Table 10-2)

TABLE 10-2. Spacecraft Subsystems. A spacecraft consists of functional groups of equipment or subsystems.

Subsystem	Principal Functions	Other Names	References
Propulsion	Provides thrust to adjust orbit and attitude, and to manage angular momentum	*Reaction Control System (RCS)*	Sec. 10.4.1, Chap. 17
Attitude Determination & Control System (ADCS)	Provides determination and control of attitude and orbit position, plus pointing of spacecraft and appendages	*Attitude Control System (ACS), Guidance, Navigation, & Control (GN&C) System, Control System*	Secs. 10.4.2, 11.1, 11.7
Communication (Comm)	Communicates with ground & other spacecraft; spacecraft tracking	*Tracking, Telemetry, & Command (TT&C)*	Secs. 10.4.3, 11.2
Command & Data Handling (C&DH)	Processes and distributes commands; processes, stores, and formats data	*Spacecraft Computer System, Spacecraft Processor*	Secs. 10.4.4, 11.3, Chap. 16
Thermal	Maintains equipment within allowed temperature ranges	*Environmental Control System*	Secs. 10.4.5, 11.5
Power	Generates, stores, regulates, and distributes electric power	*Electric Power System (EPS)*	Secs. 10.4.6, 11.4
Structures and Mechanisms	Provides support structure, booster adapter, and moving parts	*Structure Subsystem*	Secs. 10.4.7, 11.6



Communications
1. C-band antenna reflector
2. Communications receivers
3. Antenna feed horns
4. Antenna waveguide
5. Antenna tower
6. Omni antenna
7. Transponders and solid-state
8. Electrical power conditioner

Power
9. Power supply electronics
10. Battery packs
11. Solar-array panel

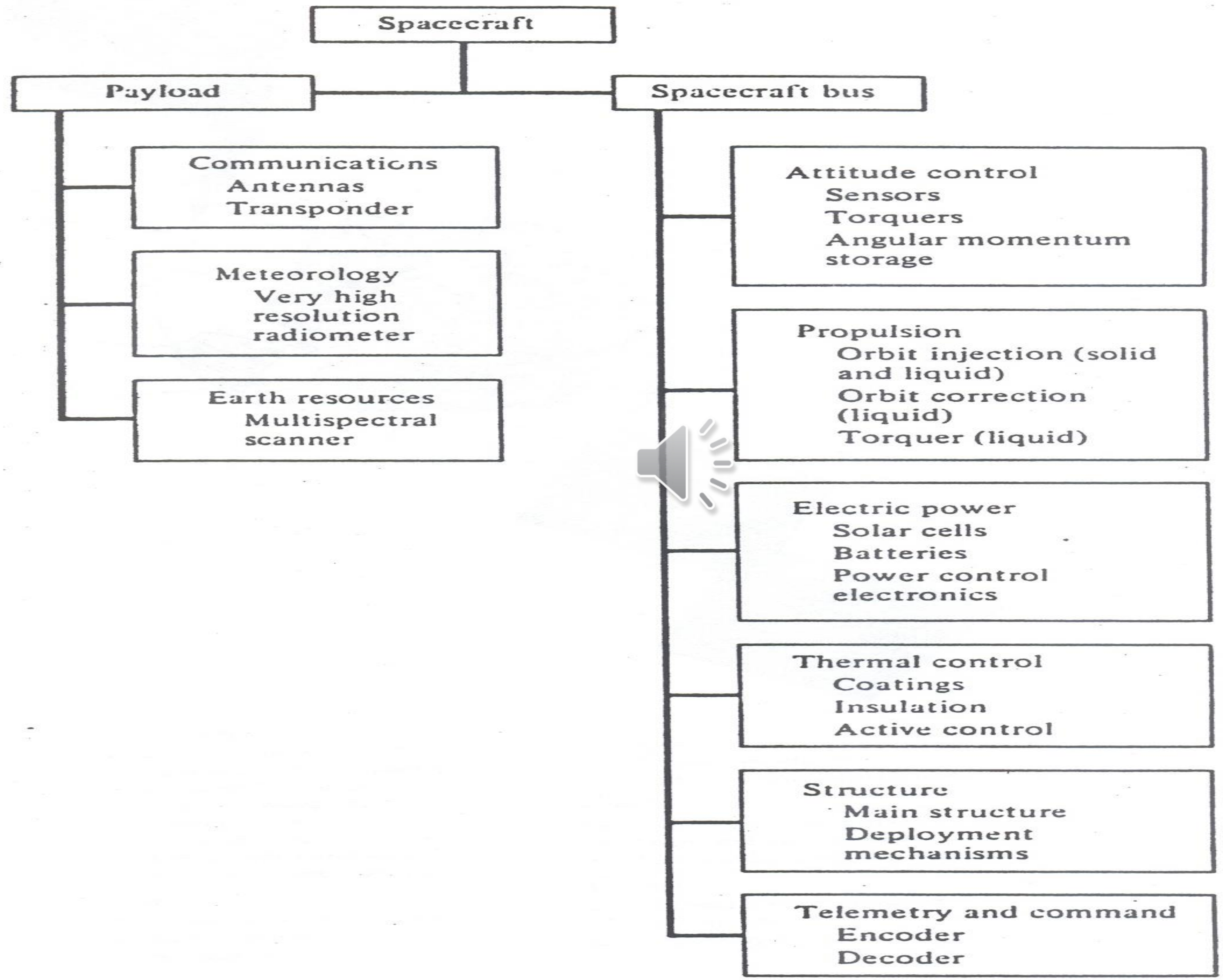
Propulsion
12. Reaction engine
13. Propulsion tanks
14. Apogee kick motor

Adacs
15. Earth scanner
16. Momentum wheel
17. Attitude control electronics

CT&T
18. Central logic processor
19. Command logic decoder
20. Transponder control electronics

Thermal
21. Thermal blanket structure
22. Central core
23. Solar-array boom

Figure 5-8 Advanced RCA Satcom. (Reprinted by permission of RCA Astro-Electronics.)



Spacecraft block diagram.

Propulsion Subsystem

- Reaction Control Subsystem (RCS)

- Provide thrust for changing the S/C velocity
- Provide torques to change its angular momentum ⇒ orbit, attitude control

- Equipment
  - Propellant
  - Tankage
  - Distribution System
  - Pressurant
  - Propellant Controls
  - Thruster (Engine)

- Sizing Parameter
  - Total Impulse
  - Number, Orientation, Thrust Level of Thruster ⇒ Engine Type

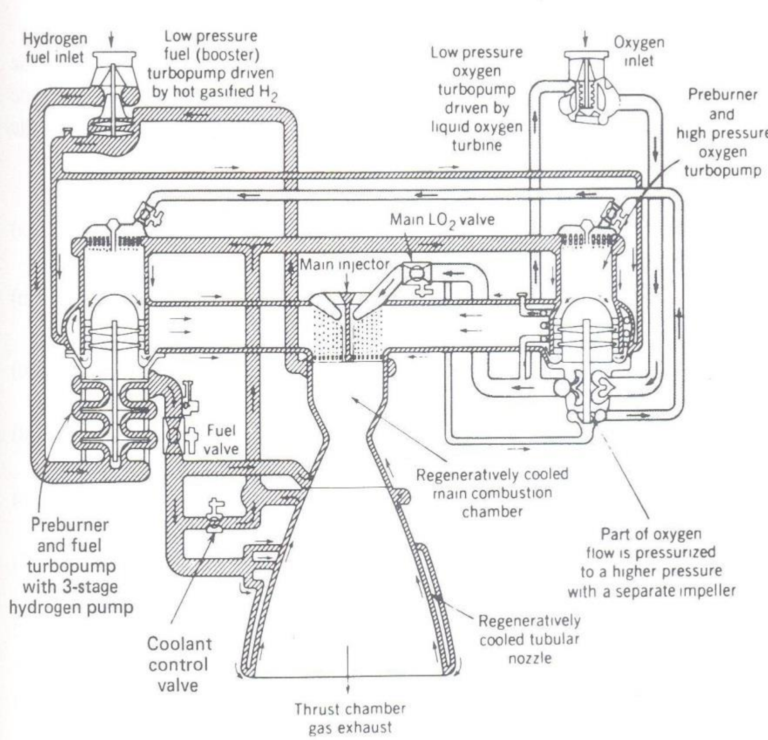
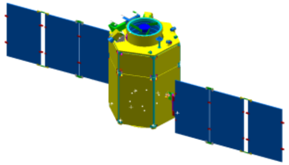


Fig. 5.12 Rocket engine diagram—preburner cycle (SSME).

SSME: Space Shuttle Main Engine



Attitude Determination and Control Subsystem (ADCS)

- Attitude Control System (ACS)
- Guidance, Navigation & Control System (GN&C)
- Control System

- Measure/control the S/C's angular orientation
- GN&C and Control System usually means control on both orientation & linear velocity (orbit)

- Simple Passive Control
  - Spin
  - Earth's Magnetic/Gravity Gradient

- Active Control
  - Sensor
    - Star, Sun, Earth, etc
  - Actuator
    - Momentum Wheel (MW)
    - Reaction Wheel (RW)
    - Contro Moment Gyro (CMG)

Attitude Determination and Control Subsystem (ADCS)

- Capability depends on :
  - Number of Body Axes
  - Appendages (Solar Arrays, Antennas …)
  - Control Accuracy
  - Speed of Response
  - Disturbance Environment

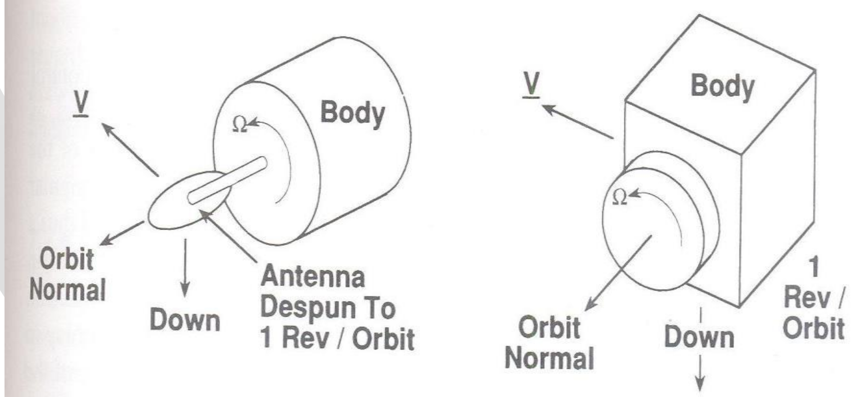


Fig. 7.17 Dual-spin spacecraft configuration concept.

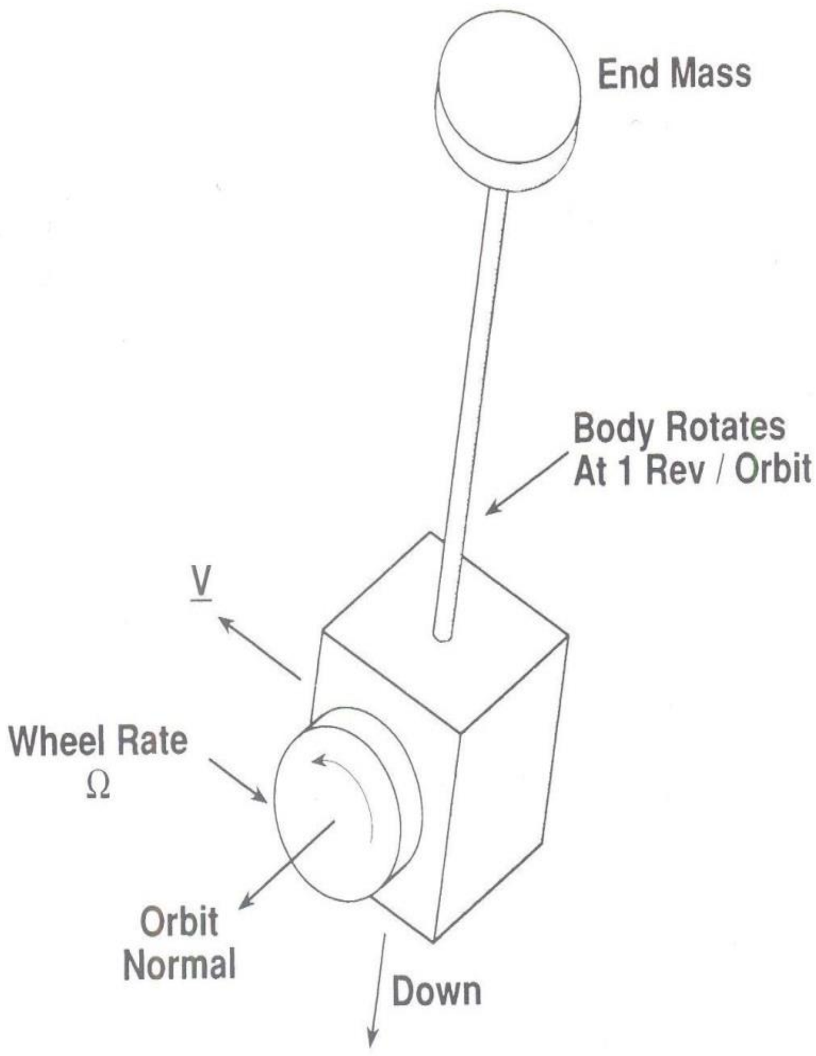


Fig. 7.14 Gravity-gradient stabilization with momentum wheel.

Communications Subsystem (Comm.)

- Tracking Telemetry & Command (TT&C)
- Link the S/C with ground or other S/C

- Equipments
  - Receiver
  - Transmitter
  - Antenna

- Size is determined by
  - Data Rate
  - Allowable Error Rate
  - Communication Path Length
  - RF Frequency

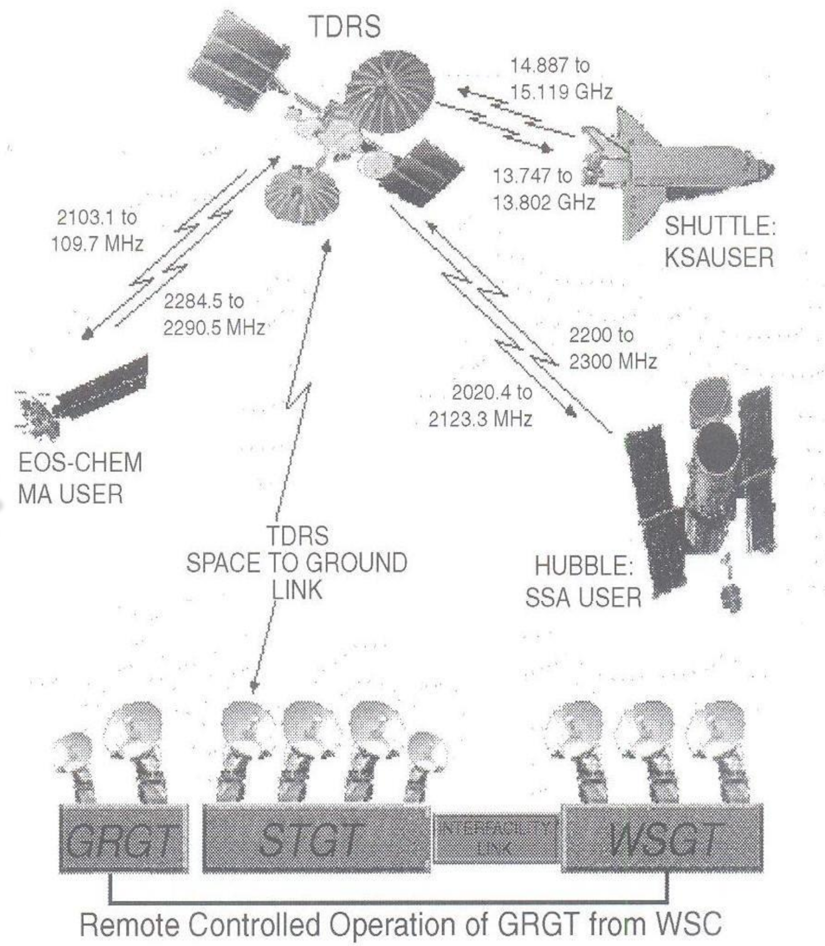


Fig. 11.20 TDRSS operations concept.

TDRSS: Tracking and Data Relay Satellite System

Command and Data Handling Subsystem (C&DH)

- S/C Computer System
- S/C Processor

- Distribute commands
- Accumulates/Stores/Formats data from S/C and payload.

- System includes
  - Central Processor (Computer)
  - Data Buses
  - Remote Interface Units
  - Data Volume
  - Rate

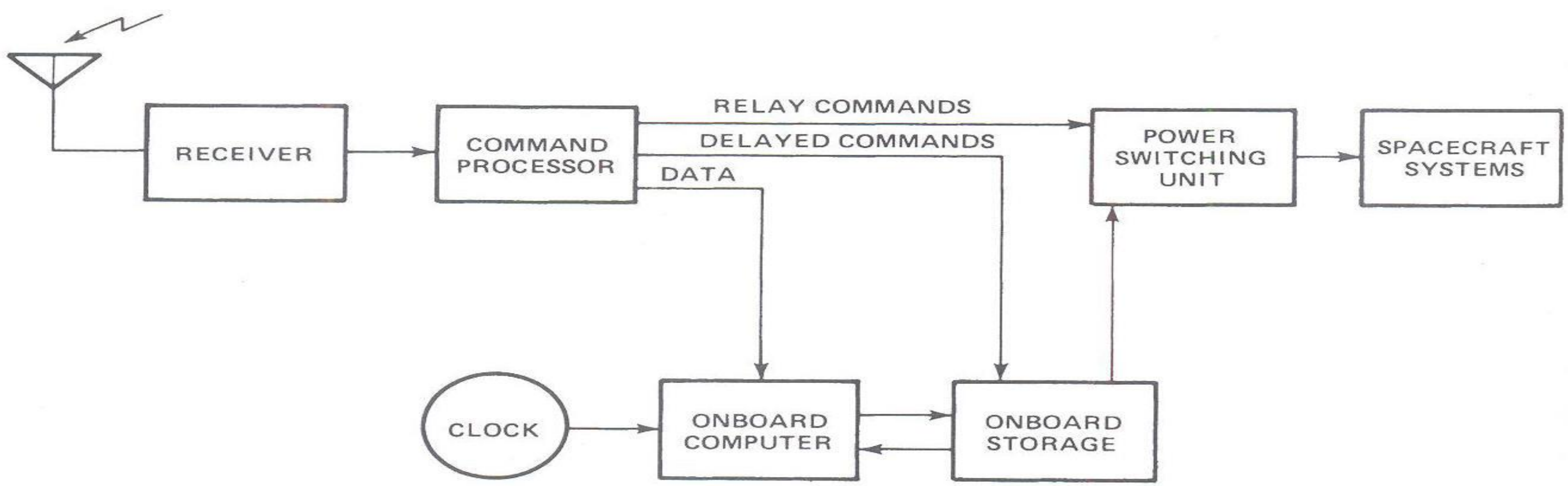


Fig. 11.1 Spacecraft command system block diagram.

Power Subsystem

- Electric power system (EPS)
- Provide electric power for the equipment on the S/C and payload.

- Consists of:
  - Power Source (Solar Cells)
  - Power Storage (Batteries)
  - Power Conversion Distribution equipment.
- Size is determined by the power needed to operate the equipment and power duty cycle

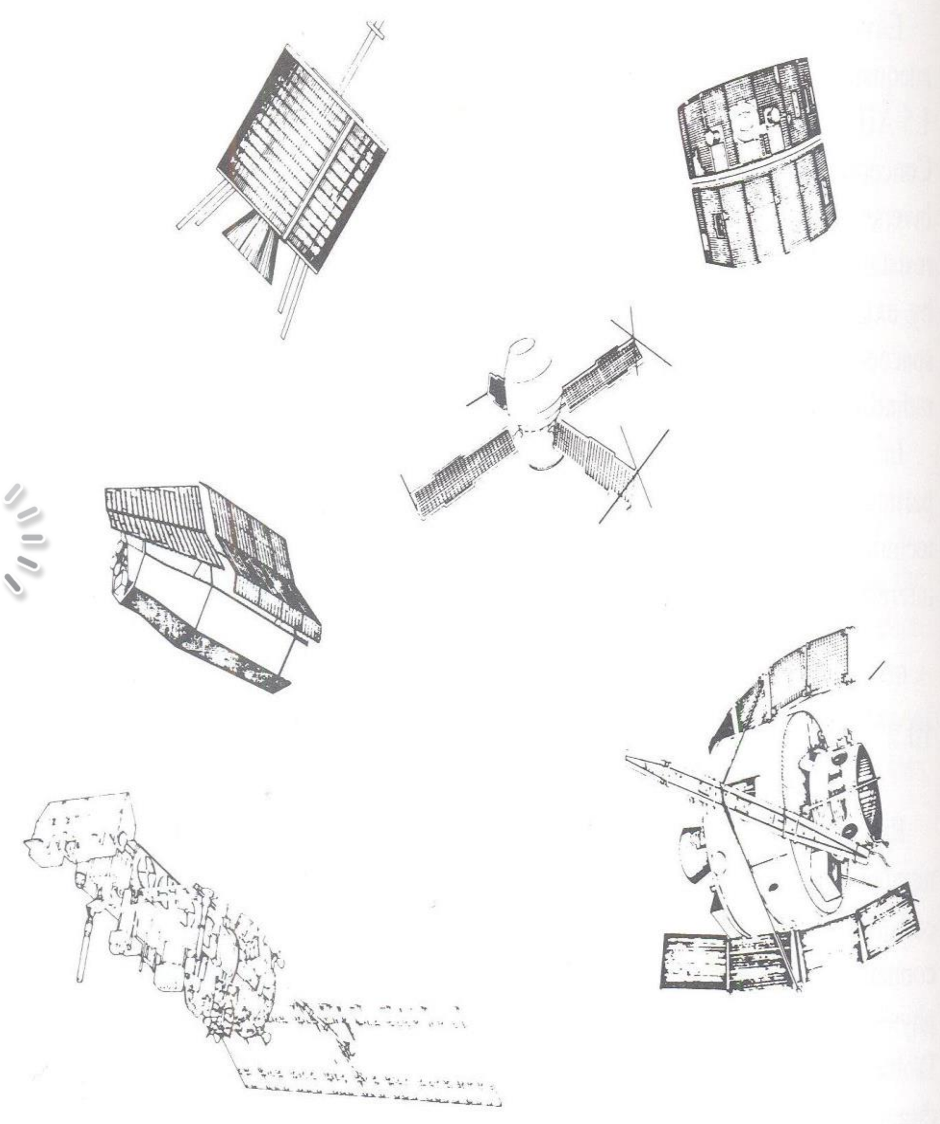


Fig. 10.5 Spacecraft solar array concepts.

Thermal Subsystem

- Thermal Control System
- Environment Control System (ECS)
- Control the S/C equipment's temperature

- Passive Method
  - Thermal Insulation
  - Coating
- Active Method
  - Electrical Heaters
  - High-Capacity, Heat-Conductors
  - Heat Pipes

- Size is determined by heat dissipation and temperatures required for equipment operation

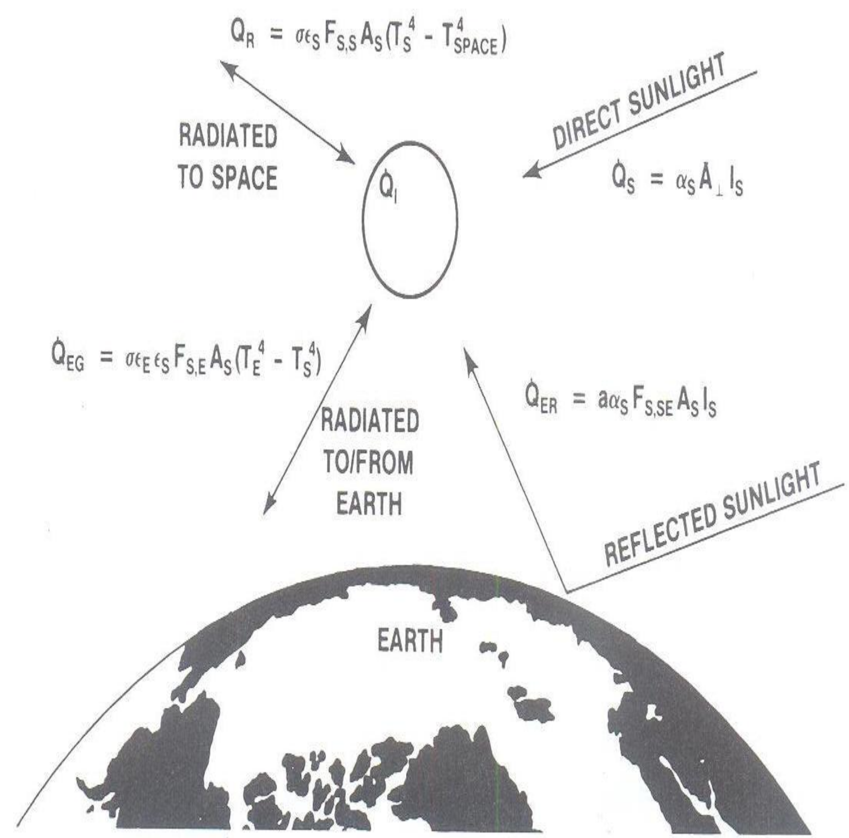


Fig. 9.7 Energy balance for an Earth orbiting spacecraft.

Structural Subsystem

- Carry/Support/Align the S/C equipment primary structure

- Primary Structure is sized by the strength or stiffness during boost
- Secondary Structure consists of deployables/supports for components

- Mechanism
  - Deployment Hinge
  - Latching Mechanisms
  - Rotating Devices
  - Solar Array Drives
  - S/C Harness-Cables

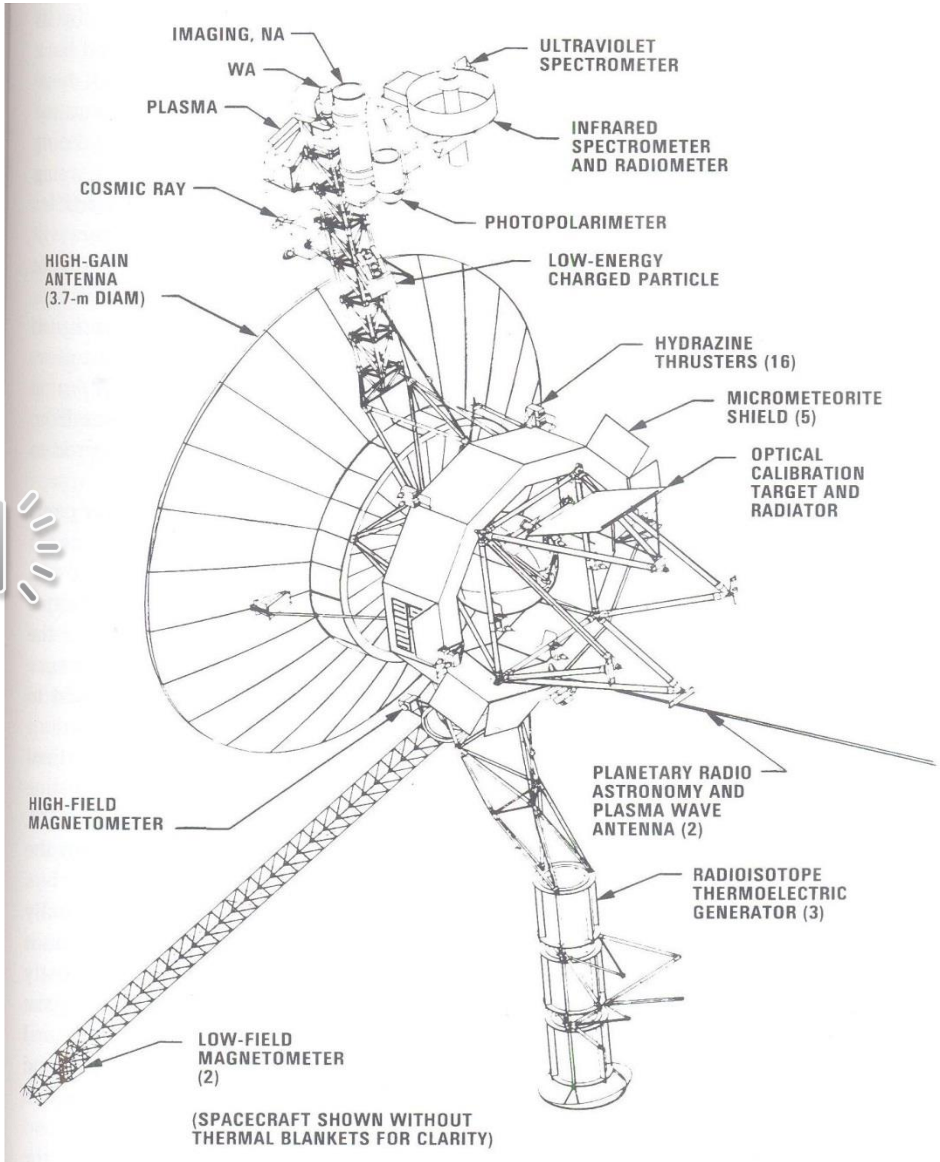


Fig. 8.1 Voyager spacecraft. (Courtesy of Jet Propulsion Laboratory.)